GH-18 · Revision R1 · Materials and quantities

Design-stage take-off from the 3D model (house_r1.json, tools/boq.py). Quantities only; wastage as stated; to be confirmed on site. The editable Excel file has every line: GH-18_R1_materials.xlsx.

Summary

1. Tiles	Wet-room walls 1376 sq ft; tiled floors 1643 sq ft (washrooms, servant room, store, balcony, terraces); 445 sq ft waterproofing
Living floors	Option A Italian marble 2104 sq ft net (+15% wastage = 2420 sq ft) | Option B 1600 × 800 vitrified 86 boxes | skirting 496 rft
2. Pipes	Drainage 76 m, water 92 m, rainwater 41 m (+ AC copper 45 m)
3. Wires	Power and lighting cable 3631 m, Cat-6 76 m, conduit 1286 m
4. Paint	Interior walls 3846 sq ft, ceilings 2535 sq ft, exterior 2693 sq ft; about 92 L interior emulsion, 50 L exterior
5. POP	False ceiling 2535 sq ft (120 gypsum boards), cove 592 rft
6. Granite	Counter tops 84 sq ft, 12 sills 47 sq ft, 24 thresholds 45 sq ft, stair 236 sq ft, flamed paving 147 sq ft
7. Timber	15 door frames (22.7 cft) + 16 shutters; pergola 36.0 cft; handrail 43 rft; veneer cladding 684 sq ft; fluted walnut 301 sq ft
8. MDF	Wardrobes 72 sheets 18 mm + 24 sheets 8 mm; other storage 73 + 30; BWP ply (wet) 15 sheets; 366 soft-close hinges
9. Lights	88 downlights (on 36 light points), 7 fans, 181 m cove LED, 9 m under-cabinet LED, 4 mirror, 10 wall, 8 garden lights

Rooms

Floor	Room	Area sq ft	Floor	Marble A sq ft	Tiles B boxes	Wall paint sq ft	Wall tiles sq ft	POP sq ft	Cove rft	Lights
GF	Kitchen	171	Italian marble	197	7	396		171	56.0	7 downlights, 2 light points, 2 exhaust, 1 heat det.
GF	Dining	149	Italian marble	171	6	240		149	30.0	8 downlights, 2 light points, 1 pendant, 1 mirror light
GF	Family Lounge	179	Italian marble	205	7	135		179	38.0	8 downlights, 2 light points, 1 wall light, 1 fan, 1 smoke det.
GF	Lobby	137	Italian marble	157	6			137	45.0	5 downlights, 2 light points, 1 smoke det.
GF	Toilet	90	Porcelain 1200×600				315	90		2 downlights, 2 light points, 1 exhaust
GF	Guest Room	202	Italian marble	232	8	432		202	57.0	6 downlights, 2 light points, 1 fan, 1 smoke det.
GF	Entrance Foyer	60	Italian marble	69	3			60	32.0	2 downlights, 2 light points
GF	Guest Lounge / Study	147	Italian marble	169	6			147	49.0	6 downlights, 2 light points, 1 fan, 1 smoke det.
GF	Under-stair Storage	51	Flamed granite			230				1 light points
FF	Master Washroom	55	Porcelain 1200×600				238	55		1 downlights, 2 light points, 1 mirror light, 1 exhaust

FF	Dresser	87	Italian marble	101	4	268		87	37.0	2 downlights, 1 light points, 1 mirror light
FF	Master Bedroom	275	Italian marble	316	11	399		275	72.0	9 downlights, 3 light points, 1 fan, 1 smoke det.
FF	Balcony	68	Deck tile							2 wall light, 2 garden light
FF	Bedroom 3 Toilet	68	Porcelain 1200×600				253	68		2 downlights, 1 light points, 1 mirror light, 1 exhaust
FF	Lobby	99	Italian marble	114	4	232		99	37.0	3 downlights, 2 light points, 1 smoke det.
FF	Bedroom 3	273	Italian marble	314	11	536		273	71.0	9 downlights, 2 light points, 1 fan, 1 smoke det.
FF	Kids Room	227	Italian marble	260	9	443		227	67.0	9 downlights, 2 light points, 1 fan, 1 smoke det.
FF	Kids Toilet	50	Porcelain 1200×600				237	50		2 downlights, 1 light points, 1 exhaust
SF	Lobby	99	Italian marble	114	4	217		99		3 downlights, 2 light points, 1 smoke det.
SF	Servant Washroom	26	Porcelain 1200×600				141	26		1 downlights, 1 light points, 1 exhaust
SF	Store	39	Porcelain 1200×600				193	39		1 downlights, 1 light points
SF	Servant Room	102	Vitrified 600×600			318		102		2 downlights, 1 light points, 1 fan, 1 smoke det.
SF	Back Terrace + Utility (covered)	430	Deck tile							2 wall light
SF	Front Terrace (open)	455	Deck tile							

1. Tiles

Floor	Room / area	Surface	Material	Net m2	Net sq ft	Wastage %	Boxes
GF	Toilet	Floor	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	8.37	90	10	7
GF	Toilet	Walls (full height to the false ceiling)	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	29.28	315	10	23
FF	Master Washroom	Floor	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	5.11	55	10	4
FF	Master Washroom	Walls (full height to the false ceiling)	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	22.1	238	10	17
FF	Balcony	Floor	Wood-look porcelain deck tile 1200 × 200 mm, R11 anti-skid (balcony, terraces)	6.3	68	10	5
FF	Bedroom 3 Toilet	Floor	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	6.36	68	10	5

FF	Bedroom 3 Toilet	Walls (full height to the false ceiling)	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	23.46	253	10	18
FF	Kids Toilet	Floor	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	4.65	50	10	4
FF	Kids Toilet	Walls (full height to the false ceiling)	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	22.02	237	10	17
SF	Servant Washroom	Floor	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	2.43	26	10	2
SF	Servant Washroom	Walls (full height to the false ceiling)	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	13.06	141	10	10
SF	Store	Floor	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	3.64	39	10	3
SF	Store	Walls (full height to the false ceiling)	Matte porcelain 1200 × 600 mm, warm grey (washrooms, floors + walls)	17.92	193	10	14
SF	Servant Room	Floor	Vitrified 600 × 600 mm, light (SF service rooms)	9.46	102	8	8
SF	Back Terrace + Utility (covered)	Floor	Wood-look porcelain deck tile 1200 × 200 mm, R11 anti-skid (balcony, terraces)	39.96	430	10	31
SF	Front Terrace (open)	Floor	Wood-look porcelain deck tile 1200 × 200 mm, R11 anti-skid (balcony, terraces)	42.27	455	10	33
GF	Deck + Pergola	Floor	Wood-look porcelain deck tile 1200 × 200 mm, R11 anti-skid (balcony, terraces)	24.1	259	10	19

2. Pipes

Kind	Dia mm	Specification	Length m	Length ft	3 m lengths (+10%)	Bends (approx.)	Runs / items
cold	25	CPVC SDR 11, 1" (25 mm)	32.9	107.8	13	9	8
cold	32	CPVC SDR 11, 1 1/4" (32 mm)	15.8	52.0	6	2	1
hot	25	CPVC SDR 11, 1" (25 mm), hot	17.5	57.3	7	5	2
pumped	32	CPVC SDR 11, 1 1/4" (32 mm), pumped riser	26.0	85.2	10	6	1
rain	100	uPVC SWR 110 mm (rainwater)	40.6	133.3	15	1	7
sewer	110	uPVC SN8 110 mm, underground	4.9	16.1	2	1	1
sewer	150	uPVC SN8 150 mm, underground	18.0	58.9	7	4	4
soil	110	uPVC SWR 110 mm, Type B, ring-fit	29.5	96.7	11	2	7
waste	50	uPVC SWR 50 mm, Type A	2.7	8.7	1	1	2
waste	75	uPVC SWR 75 mm, Type A	20.7	68.1	8	12	8
chamber		IC-2 inspection: OTS, 600×600 mm, 700 deep					1
chamber		GT-1 grease trap: OTS, 450×450 mm, 500 deep					1

chamber		CO-1 cleanout: GF lobby, 300×300 mm, 500 deep					1
chamber		CO-2 cleanout: GF lobby / foyer, 300×300 mm, 580 deep					1
chamber		IC-3 inspection: entrance court, 600×600 mm, 800 deep					1
chamber		IC-4 inspection: parking, 600×600 mm, 700 deep					1
chamber		IC-5 inspection: parking, 600×600 mm, 900 deep					1
chamber		RWH rwh pit: entrance court, 1200×1200 mm, 2000 deep					1
chamber		UGT tank: front lawn (NE), 1500×1200 mm, 1500 deep					1
refrigerant		Copper pair 1/4" + 5/8" insulated, 2 TR AC GF-DRW → SF-AC3 (+ 20 mm PVC drain, 4C x 1.5 sq mm interconnect)	11.0	36.1			1
refrigerant		Copper pair 1/4" + 5/8" insulated, 2 TR AC FF-MST → SF-AC4 (+ 20 mm PVC drain, 4C x 1.5 sq mm interconnect)	2.7	8.7			1
refrigerant		Copper pair 1/4" + 1/2" insulated, 1.5 TR AC FF-BD3 → SF-AC1 (+ 20 mm PVC drain, 4C x 1.5 sq mm interconnect)	15.7	51.7			1
refrigerant		Copper pair 1/4" + 1/2" insulated, 1.5 TR AC FF-KID → SF-AC5 (+ 20 mm PVC drain, 4C x 1.5 sq mm interconnect)	3.5	11.4			1
refrigerant		Copper pair 1/4" + 1/2" insulated, 1.5 TR AC GF-MND → SF-AC6 (+ 20 mm PVC drain, 4C x 1.5 sq mm interconnect)	7.6	25.1			1
refrigerant		Copper pair 1/4" + 1/2" insulated, 1.5 TR AC GF-GST → GF-O08 (+ 20 mm PVC drain, 4C x 1.5 sq mm interconnect)	4.6	15.2			1

3. Wires

Cable	Circuits	Route m	Cable m (all cores + 10%)	Coils (90 m) / boxes	Conduit	Note
Cat-6 U/UTP (data, from the rack)	1	69.1	76.0	1	20 mm PVC conduit 76 m	1 run per point + 10% (305 m box)
FRLS Cu 1.5 sq mm	12	501.7	1655.5	19	20 mm PVC conduit 552 m	3 cores (phase, neutral, earth) + 10%
FRLS Cu 2.5 sq mm	10	328.5	1083.9	13	20 mm PVC conduit 361 m	3 cores (phase, neutral, earth) + 10%
FRLS Cu 4 sq mm	15	164.9	544.1	7	25 mm PVC conduit 181 m	3 cores (phase, neutral, earth) + 10%
Per maker / vendor (lift, pumps, spare): use 4 sq mm until confirmed	6	105.4	347.7	4	25 mm PVC conduit 116 m	3 cores (phase, neutral, earth) + 10%
SM-1 MTR → DB-1: 4C x 16 sq mm Cu armoured (indicative)	1	23.6	26.0		32 mm conduit / buried sleeve	4-core armoured, buried, sleeved under the lobby floor; clear of the sewer
SM-2 DB-1 → DB-2: 4C x 10 sq mm Cu armoured (indicative)	1	9.5	10.5		32 mm conduit / buried sleeve	in the GF ceiling slab, up the dresser west wall

SM-3 DB-1 → DB-3: 4C x 6 sq mm Cu armoured (indicative)	1	12.1	13.4		32 mm conduit / buried sleeve	in the GF ceiling slab, then a riser in the east-wall chase (guest room / bedroom 3 / servant room)
SM-4 DB-1 → INV: 2C x 6 sq mm Cu (inverter feed)	1	16.1	17.7		32 mm conduit / buried sleeve	inverter charger feed + UPS changeover
Accessories: 6 A sockets 27, 16 A 4, USB 2, TV 6, data 2, IP65 4, switchboards 16, bedside / 2-way 13					modular	
Distribution boards: DB-1 20 circuits (24 ways with 20% spare), DB-2 14 circuits (17 ways with 20% spare), DB-3 8 circuits (10 ways with 20% spare), INV 2 circuits (3 ways with 20% spare)					DB-1 TPN	RCDs per the circuit list in services_report.txt

4. Paint

Floor	Room / area	Surface	Area m2	Area sq ft	Putty kg (2 coats)	Primer L (1 coat)	Emulsion L (2 coats)
GF	Kitchen	Interior walls	36.77	396	29.4	3.7	5.7
GF	Kitchen	Ceiling (on POP)	15.91	171		1.6	2.5
GF	Dining	Interior walls	22.3	240	17.8	2.2	3.5
GF	Dining	Ceiling (on POP)	13.8	149		1.4	2.1
GF	Family Lounge	Interior walls	12.52	135	10.0	1.3	1.9
GF	Family Lounge	Ceiling (on POP)	16.59	179		1.7	2.6
GF	Lobby	Ceiling (on POP)	12.7	137		1.3	2.0
GF	Toilet	Ceiling (on POP)	8.37	90		0.8	1.3
GF	Guest Room	Interior walls	40.17	432	32.1	4.0	6.2
GF	Guest Room	Ceiling (on POP)	18.73	202		1.9	2.9
GF	Entrance Foyer	Ceiling (on POP)	5.59	60		0.6	0.9
GF	Guest Lounge / Study	Ceiling (on POP)	13.64	147		1.4	2.1
GF	Under-stair Storage	Interior walls	21.37	230	17.1	2.1	3.3
FF	Master Washroom	Ceiling (on POP)	5.11	55		0.5	0.8
FF	Dresser	Interior walls	24.93	268	19.9	2.5	3.9
FF	Dresser	Ceiling (on POP)	8.12	87		0.8	1.3
FF	Master Bedroom	Interior walls	37.09	399	29.7	3.7	5.7
FF	Master Bedroom	Ceiling (on POP)	25.56	275		2.6	4.0
FF	Bedroom 3 Toilet	Ceiling (on POP)	6.36	68		0.6	1.0
FF	Lobby	Interior walls	21.58	232	17.3	2.2	3.3
FF	Lobby	Ceiling (on POP)	9.22	99		0.9	1.4
FF	Bedroom 3	Interior walls	49.76	536	39.8	5.0	7.7
FF	Bedroom 3	Ceiling (on POP)	25.38	273		2.5	3.9

FF	Kids Room	Interior walls	41.16	443	32.9	4.1	6.4
FF	Kids Room	Ceiling (on POP)	21.05	227		2.1	3.3
FF	Kids Toilet	Ceiling (on POP)	4.65	50		0.5	0.7
SF	Lobby	Interior walls	20.14	217	16.1	2.0	3.1
SF	Lobby	Ceiling (on POP)	9.22	99		0.9	1.4
SF	Servant Washroom	Ceiling (on POP)	2.43	26		0.2	0.4
SF	Store	Ceiling (on POP)	3.64	39		0.4	0.6
SF	Servant Room	Interior walls	29.52	318	23.6	3.0	4.6
SF	Servant Room	Ceiling (on POP)	9.46	102		0.9	1.5
ALL	Exterior façade, parapets, boundary walls	Exterior (party walls excluded)	250.22	2693		30.0	50.0

5. POP (false ceiling)

Floor	Room	Ceiling	Height mm	Area m2	Area sq ft	Gypsum boards 1800 × 1200	Perimeter channel m	Cove m
GF	Kitchen	POP	2750	15.91	171	9	17.2	17.2
GF	Dining	POP	2750	13.8	149	8	9.0	9.0
GF	Family Lounge	POP	2750	16.59	179	9	11.6	11.6
GF	Lobby	POP	2750	12.7	137	7	13.7	13.7
GF	Toilet	POP	2600	8.37	90	5	12.7	
GF	Guest Room	POP	2750	18.73	202	10	17.5	17.5
GF	Entrance Foyer	POP + walnut veneer	2750	5.59	60	3	9.7	9.7
GF	Guest Lounge / Study	POP	2750	13.64	147	7	15.0	15.0
FF	Master Washroom	POP	2600	5.11	55	3	9.4	
FF	Dresser	POP	2750	8.12	87	5	11.4	11.4
FF	Master Bedroom	POP	2750	25.56	275	14	22.0	22.0
FF	Bedroom 3 Toilet	POP	2600	6.36	68	4	10.4	
FF	Lobby	POP	2750	9.22	99	5	11.2	11.2
FF	Bedroom 3	POP	2750	25.38	273	13	21.7	21.7
FF	Kids Room	POP	2750	21.05	227	11	20.5	20.5
FF	Kids Toilet	POP	2600	4.65	50	3	9.2	
SF	Lobby	POP	2750	9.22	99	5	12.3	
SF	Servant Washroom	POP	2600	2.43	26	2	6.4	
SF	Store	POP	2600	3.64	39	2	7.6	
SF	Servant Room	POP	2750	9.46	102	5	12.8	

6. Granite & stone

Floor	Item	Description	Net m2	Net sq ft	Wastage %	Note
GF	Kitchen	Backsplash 600 mm (white marble as designed; or granite to match)	5.82	63	10	
GF	Under-stair Storage	Floor (flamed granite)	4.75	51	10	
GF	Entrance Court	Paving (flamed granite)	8.34	90	10	
GF	Foyer step	Paving (flamed granite)	0.54	6	10	
GF	Parking	Cobble setts (granite), ramp / court	7.31	79	10	
GF	Ramp (outside plot)	Cobble setts (granite), ramp / court	4.95	53	10	
GF-SF	Stair	32 treads (with 25 mm nosing), 30 mm	9.28	100	10	or Italian marble to match the lobby
GF-SF	Stair	36 risers, 18 mm	6.96	75	10	
GF-SF	Stair	2 mid-landings	5.66	61	10	
GF	GF-K01 GF-KIT	Counter top 20 mm, black granite, front edge moulded 2.02 m	1.21	13	10	
GF	GF-K02 GF-KIT	Counter top 20 mm, black granite, front edge moulded 2.60 m	1.56	17	10	
GF	GF-K03 GF-KIT	Counter top 20 mm, black granite, front edge moulded 2.99 m; hob cut-out	1.79	19	10	
GF	GF-K10 GF-KIT	Counter top 20 mm, black granite, front edge moulded 2.10 m; sink cut-out	1.26	14	10	
FF	FF-L01 FF-LOB	Counter top 20 mm, black granite, front edge moulded 1.50 m	0.9	10	10	
SF	SF-U02 SF-TRB	Counter top 20 mm, black granite, front edge moulded 1.00 m; sink cut-out	0.6	6	10	
SF	SF-U03 SF-TRB	Counter top 20 mm, black granite, front edge moulded 1.10 m	0.5	5	10	
GF	GF-GL4	Window sill 1460 × 200 mm, 18 mm	0.292	3.1	10	
GF	GF-GL5	Door threshold 5700 × 150 mm, 18 mm	0.855	9.2	10	
GF	GF-GL3	Window sill 1150 × 350 mm, 18 mm	0.403	4.3	10	
GF	GF-KW1	Window sill 500 × 350 mm, 18 mm	0.175	1.9	10	
GF	GF-D3a	Door threshold 800 × 150 mm, 18 mm	0.12	1.3	10	
GF	GF-D2a	Door threshold 1000 × 150 mm, 18 mm	0.15	1.6	10	
GF	GF-GL1	Window sill 2400 × 200 mm, 18 mm	0.48	5.2	10	
GF	GF-KD1	Door threshold 1000 × 150 mm, 18 mm	0.15	1.6	10	
GF	GF-D3b	Door threshold 800 × 180 mm, 18 mm	0.144	1.6	10	
GF	GF-D2b	Door threshold 1000 × 180 mm, 18 mm	0.18	1.9	10	
GF	GF-D1	Door threshold 1450 × 180 mm, 18 mm	0.261	2.8	10	
GF	GF-GL2	Window sill 2770 × 230 mm, 18 mm	0.637	6.9	10	
GF	GF-GD1	Door threshold 1200 × 60 mm, 18 mm	0.072	0.8	10	
GF	GF-GL6d	Door threshold 700 × 10 mm, 18 mm	0.007	0.1	10	
FF	FF-GL4	Window sill 1460 × 200 mm, 18 mm	0.292	3.1	10	
FF	FF-MW1	Window sill 500 × 350 mm, 18 mm	0.175	1.9	10	

FF	FF-GL3	Window sill 1150 × 350 mm, 18 mm	0.403	4.3	10	
FF	FF-GL7	Door threshold 3500 × 150 mm, 18 mm	0.525	5.7	10	
FF	FF-D2a	Door threshold 1000 × 150 mm, 18 mm	0.15	1.6	10	
FF	FF-D3a	Door threshold 800 × 150 mm, 18 mm	0.12	1.3	10	
FF	FF-D2b	Door threshold 1000 × 180 mm, 18 mm	0.18	1.9	10	
FF	FF-D2c	Door threshold 1000 × 150 mm, 18 mm	0.15	1.6	10	
FF	FF-D3b	Door threshold 800 × 150 mm, 18 mm	0.12	1.3	10	
FF	FF-D2d	Door threshold 1000 × 150 mm, 18 mm	0.15	1.6	10	
FF	FF-D3c	Door threshold 800 × 150 mm, 18 mm	0.12	1.3	10	
FF	FF-GL1	Window sill 2400 × 200 mm, 18 mm	0.48	5.2	10	
FF	FF-GL10	Window sill 1000 × 350 mm, 18 mm	0.35	3.8	10	
FF	FF-GL6ad	Door threshold 700 × 10 mm, 18 mm	0.007	0.1	10	
FF	FF-GL6bd	Door threshold 700 × 10 mm, 18 mm	0.007	0.1	10	
SF	SF-PD1a	Door threshold 1000 × 180 mm, 18 mm	0.18	1.9	10	
SF	SF-PW1	Window sill 1200 × 200 mm, 18 mm	0.24	2.6	10	
SF	SF-PD1b	Door threshold 1000 × 150 mm, 18 mm	0.15	1.6	10	
SF	SF-PD2a	Door threshold 800 × 150 mm, 18 mm	0.12	1.3	10	
SF	SF-PD2b	Door threshold 800 × 150 mm, 18 mm	0.12	1.3	10	
SF	SF-PD2c	Door threshold 800 × 150 mm, 18 mm	0.12	1.3	10	
SF	SF-GL3	Window sill 1150 × 350 mm, 18 mm	0.403	4.3	10	

7. Timber & veneer

Floor	Item	Description	Unit	Qty	sq ft	Note
GF	Family Lounge	Fluted walnut feature wall (veneered MDF flutes on ply)	m2	6.32	68	
GF	Lobby	Walnut veneer wall cladding (PDF schedule)	m2	23.99	258	
GF	Entrance Foyer	Walnut veneer wall cladding (PDF schedule)	m2	6.99	75	
GF	Entrance Foyer	Fluted walnut feature wall (veneered MDF flutes on ply)	m2	8.01	86	
GF	Entrance Foyer	Walnut veneer ceiling accent (with the POP)	m2	1.68	18	about 30% of the ceiling
GF	Guest Lounge / Study	Walnut veneer wall cladding (PDF schedule)	m2	32.55	350	
FF	Master Bedroom	Fluted walnut feature wall (veneered MDF flutes on ply)	m2	10.63	114	
GF	GF-D3a	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 800 × 2400 × 35 mm flush, veneer both faces	frame cft	1.48		5.60 m frame
GF	GF-D2a	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 1000 × 2400 × 35 mm flush, veneer both faces	frame cft	1.54		5.80 m frame

GF	GF-D3b	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 800 × 2400 × 35 mm flush, veneer both faces	frame cft	1.48		5.60 m frame
GF	GF-D2b	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 1000 × 2400 × 35 mm flush, veneer both faces	frame cft	1.54		5.80 m frame
GF	GF-D1	Main door: frame 125 × 60 mm teak, 2 shutter(s) 725 × 2400 × 40 mm solid teak panelled	frame cft	1.66		6.25 m frame
FF	FF-D2a	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 1000 × 2400 × 35 mm flush, veneer both faces	frame cft	1.54		5.80 m frame
FF	FF-D3a	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 800 × 2400 × 35 mm flush, veneer both faces	frame cft	1.48		5.60 m frame
FF	FF-D2b	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 1000 × 2400 × 35 mm flush, veneer both faces	frame cft	1.54		5.80 m frame
FF	FF-D2c	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 1000 × 2400 × 35 mm flush, veneer both faces	frame cft	1.54		5.80 m frame
FF	FF-D3b	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 800 × 2400 × 35 mm flush, veneer both faces	frame cft	1.48		5.60 m frame
FF	FF-D2d	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 1000 × 2400 × 35 mm flush, veneer both faces	frame cft	1.54		5.80 m frame
FF	FF-D3c	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 800 × 2400 × 35 mm flush, veneer both faces	frame cft	1.48		5.60 m frame
SF	SF-PD2a	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 800 × 2400 × 35 mm flush, veneer both faces	frame cft	1.48		5.60 m frame
SF	SF-PD2b	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 800 × 2400 × 35 mm flush, veneer both faces	frame cft	1.48		5.60 m frame
SF	SF-PD2c	Door: frame 125 × 60 mm sal / hardwood, 1 shutter(s) 800 × 2400 × 35 mm flush, veneer both faces	frame cft	1.48		5.60 m frame
GF	Pergola	4 posts 150×150×2.7 m, beam 100×250×6.7 m, ledger 100×200×6.8 m, 15 rafters 60×200×2.63 m	cft	36.0		hardwood, walnut stain, outdoor PU
GF-SF	Stair handrail	Walnut handrail 60 × 50 mm on the spine	m	13.0	42.6	
GF	GF-F25 (mandir)	Fluted cabinet 1.1 × 0.47 × 0.45 m (veneered MDF) + marble back panel 1.1 × 2.4 m (stone)	nos	1		
GF	GF-O12 (tulsi backdrop)	Fluted walnut cladding, exterior grade, PU	m2	3.04	33	

8. MDF board (wardrobes and storage)

Floor	Id	Unit	Room	Size w x d x h mm	Board	Carcass m2	Shutters m2	Back 8 mm m2	Sheets 18/19 mm (8×4, +15%)	Sheets 8 mm	Shutters (desks / dressers: drawers)	Soft-close hinges	Edge band m	Laminate m2 (shutter faces)

GF	GF-K01	Kitchen / pantry base unit	GF-KIT	2020 x 600 x 900	MDF 18 mm	5.68	1.72	1.72	3	1	4	8	20.3	1.89
GF	GF-K02	Kitchen / pantry base unit	GF-KIT	2600 x 600 x 900	MDF 18 mm	6.72	2.21	2.21	4	1	5	10	24.9	2.43
GF	GF-K03	Kitchen / pantry base unit	GF-KIT	2986 x 600 x 900	MDF 18 mm	7.92	2.54	2.54	5	1	6	12	29.4	2.79
GF	GF-K05	Wall cabinet	GF-KIT	2986 x 350 x 750	MDF 18 mm	4.45	2.24	2.24	3	1	6	12	27.7	2.46
GF	GF-K09	Wall cabinet	GF-KIT	2600 x 350 x 750	MDF 18 mm	3.78	1.95	1.95	3	1	5	10	23.5	2.15
GF	GF-F21	Shoe console	GF-FOY	1800 x 450 x 900	MDF 18 mm	3.65	1.62	1.62	3	1	4	8	18.9	1.78
GF	GF-G04	Wardrobe	GF-GST	3970 x 600 x 2700	MDF 18 mm	24.01	10.72	10.72	14	5	7	28	85.8	11.79
GF	GF-T03	Vanity	GF-TLT	900 x 600 x 850	BWP ply 19 mm (wet)	2.16	0.41	0.41	1	1	2	4	7.2	0.45
GF	GF-F04	TV unit	GF-DRW	1800 x 400 x 450	MDF 18 mm	2.7	0.81	0.81	2	1	4	8	13.9	0.89
GF	GF-F19	Crockery unit	GF-DRW	1400 x 450 x 2100	MDF 18 mm	6.62	2.94	2.94	4	2	3	12	30.1	3.23
GF	GF-F24	Vanity	GF-DIN	700 x 450 x 850	BWP ply 19 mm (wet)	1.35	0.32	0.32	1	1	2	4	6.2	0.35
GF	GF-K10	Kitchen / pantry base unit	GF-KIT	2100 x 600 x 900	BWP ply 19 mm (wet)	5.82	1.78	1.78	3	1	4	8	20.7	1.96
GF	GF-K13	Wall cabinet	GF-KIT	2100 x 350 x 700	MDF 18 mm	3.18	1.47	1.47	2	1	4	8	18.9	1.62
GF	GF-M01	Study desk	GF-MND	1400 x 600 x 750	MDF 18 mm	3.87	0.28	1.05	2	1	0	0	9.6	0.31

GF	GF-M04	TV unit	GF-MND	1500 x 400 x 450	MDF 18 mm	2.34	0.68	0.68	2	1	3	6	11.5	0.74
GF	GF-J01	Tall storage	GF-JAN	800 x 600 x 2250	MDF 18 mm	5.58	1.8	1.8	3	1	2	8	19.9	1.98
GF	GF-J02	Tall storage	GF-JAN	800 x 600 x 1600	MDF 18 mm	3.36	1.28	1.28	2	1	2	6	13.6	1.41
GF	GF-J04	Tall storage	GF-JAN	1100 x 600 x 1400	MDF 18 mm	4.5	1.54	1.54	3	1	2	6	15.3	1.69
GF	GF-K04	Kitchen tall unit	GF-KIT	800 x 600 x 2250	MDF 18 mm	5.58	1.8	1.8	3	1	2	8	19.9	1.98
GF	GF-K12	Tall storage	GF-KIT	400 x 600 x 2250	MDF 18 mm	4.14	0.9	0.9	2	1	1	4	12.2	0.99
FF	FF-M04	Wardrobe	FF-MST	2056 x 600 x 2700	MDF 18 mm	13.88	5.55	5.55	8	3	4	16	48.8	6.11
FF	FF-M05	Wardrobe	FF-MST	1127 x 600 x 2700	MDF 18 mm	8.92	3.04	3.04	5	2	2	8	27.9	3.35
FF	FF-B01	Wardrobe	FF-BD3	2038 x 600 x 2700	MDF 18 mm	13.82	5.5	5.5	8	3	4	16	48.7	6.05
FF	FF-B11	Wardrobe	FF-BD3	897 x 600 x 2700	MDF 18 mm	6.47	2.42	2.42	4	1	2	8	23.4	2.66
FF	FF-B05	Dresser	FF-BD3	1200 x 450 x 750	MDF 18 mm	2.63	0.24	0.9	2	1	0	0	8.7	0.26
FF	FF-B08	Storage bench	FF-BD3	2400 x 450 x 450	MDF 18 mm	4.05	1.08	1.08	2	1	5	10	18.3	1.19
FF	FF-T22	Vanity	FF-TL3	900 x 600 x 850	BWP ply 19 mm (wet)	2.16	0.41	0.41	1	1	2	4	7.2	0.45
FF	FF-K01	Wardrobe	FF-KID	4550 x 553 x 2700	MDF 18 mm	25.55	12.29	12.29	15	5	8	32	98.5	13.51

FF	FF-K06	Study desk	FF-KID	1600 x 600 x 750	MDF 18 mm	4.23	0.32	1.2	2	1	0	0	10.7	0.35
FF	FF-L01	Kitchen / pantry base unit	FF-LOB	1500 x 600 x 900	MDF 18 mm	4.23	1.27	1.27	3	1	3	6	15.2	1.4
FF	FF-L05	Wall cabinet	FF-LOB	1500 x 350 x 750	MDF 18 mm	2.36	1.12	1.12	2	1	3	6	14.2	1.24
FF	FF-T11	Vanity	FF-KTL	1065 x 600 x 850	BWP ply 19 mm (wet)	2.73	0.48	0.48	2	1	2	4	8.5	0.53
FF	FF-K07	Tall storage	FF-KID	1200 x 450 x 2400	MDF 18 mm	6.48	2.88	2.88	4	2	3	12	31.2	3.17
FF	FF-T04	Vanity	FF-MTL	1000 x 550 x 850	BWP ply 19 mm (wet)	2.39	0.45	0.45	2	1	2	4	8.1	0.5
FF	FF-D01	Dressing table	FF-DRS	1100 x 450 x 750	MDF 18 mm	2.5	0.22	0.83	2	1	0	0	8.2	0.24
FF	FF-D02	Wardrobe	FF-DRS	1700 x 600 x 2700	MDF 18 mm	10.98	4.59	4.59	7	2	3	12	37.9	5.05
FF	FF-D03	Wardrobe	FF-DRS	560 x 600 x 2700	MDF 18 mm	5.26	1.51	1.51	3	1	1	4	15.3	1.66
FF	FF-D05	Wardrobe	FF-DRS	800 x 600 x 2700	MDF 18 mm	6.12	2.16	2.16	4	1	2	8	22.6	2.38
FF	FF-D06	Storage bench	FF-DRS	1460 x 450 x 450	MDF 18 mm	2.58	0.66	0.66	2	1	3	6	11.3	0.72
FF	FF-M06	TV unit	FF-MST	1800 x 400 x 450	MDF 18 mm	2.7	0.81	0.81	2	1	4	8	13.9	0.89
SF	SF-S02	Wardrobe	SF-SRV	1000 x 600 x 2100	MDF 18 mm	7.38	2.1	2.1	4	1	2	8	22.7	2.31
SF	SF-S03	Desk	SF-SRV	900 x 500 x 750	MDF 18 mm	2.1	0.18	0.68	1	1	0	0	6.4	0.2

SF	SF-U02	Utility Counter	SF-TRB	1000 x 600 x 900	BWP ply 19 mm (wet)	3.33	0.85	0.85	2	1	2	4	11.0	0.94
SF	SF-U04	Tall storage	SF-TRB	600 x 600 x 2100	BWP ply 19 mm (wet)	4.68	1.26	1.26	3	1	2	8	17.4	1.39
SF	SF-S11	Tall storage	SF-WSH	1400 x 600 x 2400	MDF 18 mm	9.36	3.36	3.36	5	2	3	12	32.8	3.7

9. Lights & fittings

Fitting	Qty	Unit	Where
Downlight fixtures, 10 W slim COB 3000 K, cut-out 75 mm, on the ~1.4 m grid shown in the 3D (each light point above switches a group)	88	nos	GF-KIT 7, GF-DIN 8, GF-DRW 8, GF-LOB 5, GF-TLT 2, GF-GST 6, GF-FOY 2, GF-MND 6, FF-MTL 1, FF-DRS 2, FF-MST 9, FF-TL3 2, FF-LOB 3, FF-BD3 9, FF-KID 9, FF-KTL 2, SF-LOB 3, SF-SWR 1, SF-WSH 1, SF-SRV 2
Light points (switched groups, per the electrical sheet); wet rooms IP44	36	nos	FF-BD3 2, FF-DRS 1, FF-KID 2, FF-KTL 1, FF-LOB 2, FF-MST 3, FF-MTL 2, FF-TL3 1, GF-DIN 2, GF-DRW 2, GF-FOY 2, GF-GST 2, GF-JAN 1, GF-KIT 2, GF-LOB 2, GF-MND 2, GF-TLT 2, SF-LOB 2, SF-SRV 1, SF-SWR 1, SF-WSH 1
Statement pendant (dining)	1	nos	GF-DIN 1
Mirror light 600 mm, IP44	4	nos	FF-DRS 1, FF-MTL 1, FF-TL3 1, GF-DIN 1
Wall / niche light (IP65 outdoors)	10	nos	FF-BAL 2, GF-COURT 2, GF-DECK 2, GF-DRW 1, GF-LAWN-B 1, SF-TRB 2
Garden bollard, 7 W, IP65	8	nos	FF-BAL 2, GF-COURT 2, GF-LAWN-B 4
Ceiling fan, BLDC 1200 mm	7	nos	FF-BD3 1, FF-KID 1, FF-MST 1, GF-DRW 1, GF-GST 1, GF-MND 1, SF-SRV 1
Exhaust fan 200 mm	7	nos	FF-KTL 1, FF-MTL 1, FF-TL3 1, GF-KIT 2, GF-TLT 1, SF-SWR 1
Smoke detector	10	nos	FF-BD3 1, FF-KID 1, FF-LOB 1, FF-MST 1, GF-DRW 1, GF-GST 1, GF-LOB 1, GF-MND 1, SF-LOB 1, SF-SRV 1
Heat detector (kitchen)	1	nos	GF-KIT 1
LED strip under wall cabinets, 10 W/m 3000 K, aluminium profile	9.2	m	4 runs
Cove LED strip, 12 W/m 2700 K, in POP coves	180.6	m	+ 37 drivers (1 per 5 m)

How it is measured

How the quantities are measured (all from house_r1.json):

- Floors: room polygons (net). Option A Italian marble +15% wastage; Option B 1600 × 800 vitrified +8%, 2 tiles per box. Skirting = walled perimeter minus doors.

- Tiles: washroom walls full height to the 2.6 m false ceiling, minus doors and windows, +10%. Waterproofing = floor + 300 mm up the walls (more in the shower: 1.8 m).

- Paint: interior walls to the false ceiling minus openings, tiles, veneer cladding, fluted walls and the kitchen backsplash; ceilings = POP area; exterior = non-party external walls x storey height, parapets and boundary walls (both faces where both are exposed). Coverage rates are indicative.

- POP: ceilings of type POP (gypsum board 12.5 mm on GI framing, POP-finished), boards +10%; cove length = room perimeter where the ceiling has a cove.

- Granite: counter tops (20 mm black granite, moulded front), sills on every window (wall + 50 mm), thresholds on every door, stair treads / risers / landings (or marble to match the lobby), flamed granite paving. +10% wastage.

- Timber: door frames 125 × 60 mm (teak for the main door), shutters 35 mm flush with veneer (main door 40 mm teak); pergola hardwood; handrail; walnut veneer cladding (lobby, foyer, guest lounge, per the PDF schedule) and fluted walnut feature walls.
- MDF: each wardrobe / storage unit = carcass (sides, top, bottom, a vertical every 900 mm, 4 shelves if tall) + shutters (18 mm) + back (8 mm); +15% on 8 × 4 ft sheets. BWP ply for the vanities, the utility counter, the kitchen sink base and units on the terrace (owner: MDF elsewhere).
- Lights: counts from the electrical points; cove LED = cove length; under-cabinet LED = the cabinet runs.
- Pipes: lengths along the modelled routes and stacks, 3 m lengths +10%; bends counted where a route changes direction (tees and couplers extra).
- Wires: route length of each circuit x 3 cores (phase, neutral, earth) +10%; 90 m coils; sub-main sizes indicative (electrician to size).

Not included: labour, adhesives other than noted, hardware beyond hinges, sanitaryware, glass, aluminium windows, the lift, ACs and appliances.